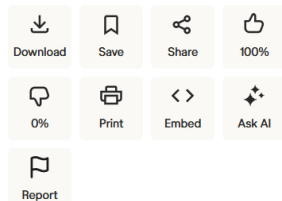


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Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**

		Module – 1	M	RBTL	COs
Q1	(a)	What is Natural Language Processing? Outline the Challenges of NLP.	08M	L2	CO1
	(b)	Consider the following Corpus of three sentences • The Arabian Knights • These are the fairy tales of the east • The stories of the Arabian Knights are translated in many languages. Calculate P for the sentence “The Arabian Knights are the fairy tales of the east” assuming a bi-gram language model.	08M	L3	CO1
	(c)	Outline Processing Indian Languages.	04M	L2	CO1
OR					
Q2	(a)	Explain Layered representation of Panninian Grammar for Indian Languages. Explain Karakas theory with example.	10M	L2	CO1
	(b)	Summarize Phrase Structure Grammar and transformational Grammar with examples. Represent sentences in s-surface structure for: • The Snatcher is caught by the police. Explain Statistical and n gram model with example.	10M	L2	CO1
Module – 2					
Q3	(a)	Explain DFA and NFA with structural representations. List the properties of Finite Automation.	08M	L2	CO2
	(b)	What is morphological Parsing? Explain two step morphological parser with a neat diagram and examples.	08M	L2	CO2
	(c)	What is Minimum edit distance? Compute the minimum edit distance between tutor and tumour.	04M	L3	CO2
OR					
Q4	(a)	Outline Grammar rules on: (i) Phrase level Constructors (ii) Sentence level constructors (iii) Agreement (iv) Coordination	08M	L2	CO2
	(b)	Explain Top down Parsing and bottom Up parsing.	08M	L2	CO2
	(c)	Summarize on CYK Parsing.	04M	L2	CO2
Module – 3					
Q5	(a)	Explain Naïve Bayes Classifier in detail.	08M	L2	CO3

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	(b)	Consider the following miniature training and test documents simplified worked example:	08M	L3	CO3																					
		<table border="1"> <thead> <tr> <th></th><th>Cat</th><th>Documents</th></tr> </thead> <tbody> <tr> <td>Training</td><td>-</td><td>just plain boring</td></tr> <tr> <td></td><td>-</td><td>entirely predictable and lacks energy</td></tr> <tr> <td></td><td>-</td><td>no surprises and very few laughs</td></tr> <tr> <td></td><td>+</td><td>very powerful</td></tr> <tr> <td></td><td>+</td><td>the most fun film of the summer</td></tr> <tr> <td>Test</td><td>?</td><td>predictable with no fun</td></tr> </tbody> </table> Calculate the probability of positive and negative class set of words for test set based on training set of sentences.		Cat	Documents	Training	-	just plain boring		-	entirely predictable and lacks energy		-	no surprises and very few laughs		+	very powerful		+	the most fun film of the summer	Test	?	predictable with no fun			
	Cat	Documents																								
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	+	very powerful																								
	+	the most fun film of the summer																								
Test	?	predictable with no fun																								
	(c)	Briefly explain about Naïve Bayes for other Text Classification Tasks.	04M	L2	CO3																					
		OR																								
Q.6	(a)	What is Binary Naïve Bayes? Interpret the example of binarization for Binary Naïve Bayes algorithm with NB counts and Binary counts.	10M	L2	CO3																					
	(b)	Explain on Naïve Bayes as a Language Model with example. Consider Naïve Bayes Model with classes positive (+) and negative (-) and following model parameters:	10M	L2	CO3																					
		<table border="1"> <thead> <tr> <th>w</th><th>P(w +)</th><th>P(w -)</th></tr> </thead> <tbody> <tr> <td>1</td><td>0.1</td><td>0.2</td></tr> </tbody> </table>	w	P(w +)	P(w -)	1	0.1	0.2																		
w	P(w +)	P(w -)																								
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		love 0.1 0.001 this 0.01 0.01 fun 0.05 0.005 film 0.1 0.1			
		Calculate Probability of positive and negative set of words in a sentence.			
Module – 4					
Q.7	(a)	Explain on Design features of Information Retrieval Systems.	10M	L2	CO4
	(b)	Outline the following: • Cluster Model • Fuzzy Model • LSTM Model • Zipf's Law	10M	L2	CO4
OR					
Q.8	(a)	Explain WordNet and FrameNet with applications.	10M	L2	CO4
	(b)	Summarize Parts of Speech Tagger and Research Corpora with illustrations.	10M	L2	CO4
Module – 5					
Q.9	(a)	Explain Language Divergences and Typology	10M	L2	CO5
	(b)	Explain Machine Translation using Encoder-Decoder with model of Beam Search Decoder.	10M	L2	CO5
OR					
Q.10	(a)	Explain translation in Low Resource situations.	10M	L2	CO5
	(b)	Explain MT evaluation in detail.	10M	L2	CO5

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